

Project Report: Customer Behavior and Sales Analysis

1. Executive Summary

- **Project Overview:** This project evaluates customer purchasing habits, product performance, and subscription metrics to uncover revenue growth opportunities and optimize retail strategies.
- **Key Metrics:**
 - Total Customers: **3,9K** (3,900 unique customers)
 - Total Sales/Revenue: **\$233K** (\$232,881 total purchase volume)
 - Average Purchase Amount: **\$59.76**
- **Primary Findings:** The customer base is heavily skewed toward non-subscribers, indicating significant untapped potential for loyalty program conversion. Revenue is largely driven by core categories such as Clothing and Accessories, with Young Adults and Middle-aged segments acting as primary revenue drivers.

2. Objectives

- **Demographics:** Understand customer distribution across age, gender, and geographical locations.
- **Product Performance:** Identify top-performing product categories and high-rated items.
- **Seasonal Trends:** Analyze purchase behavior across different seasons.
- **Subscription Patterns:** Evaluate the behavior, spending, and volume differences between subscribers and non-subscribers.

3. Data & Methodology

- **Data Preparation & Cleaning:** Loaded the raw dataset (customer_shopping_behavior.csv) using Python (pandas), handled missing review ratings via category median imputation, standardized column headers into snake_case, and performed feature engineering (such as creating age_group and purchase frequency intervals).
- **Aggregations & Queries:** Loaded the cleaned dataset into MySQL via SQLAlchemy to run structured SQL queries for revenue breakdowns, discount efficiencies, and customer segmentation.
- **Dashboard & Visualization:** Built an interactive Power BI dashboard to synthesize KPIs, category shares, and demographic filters for stakeholder review.

4. Key Insights

- **Subscription Gap:** 73% of customers are non-subscribers (2,847 out of 3,900 customers), presenting a major growth and retention opportunity.
- **Revenue Drivers:** Clothing and Accessories generate the highest overall revenue and

transaction volume across the product lines.

- **High-Spending Demographics:** Young Adults and Middle-aged groups exhibit the highest cumulative spend and frequency compared to other age brackets.

5. Strategic Recommendations

- **Boost Subscriptions:** Launch targeted marketing campaigns and tiered perks (e.g., exclusive discounts or free express shipping) to convert the 73% non-subscriber base into loyal members.
- **Optimize Inventory:** Allocate more inventory space and promotional budgets toward top-performing categories like Clothing and Accessories to maximize stock turnover.
- **Target Key Demographics:** Tailor digital advertisements and seasonal promotions specifically toward Young Adults and Middle-aged segments, who form the core purchasing power.